VULNIQ

346,232 vulnerabilities.

Which ones deserve attention first?

Machine-learning based vulnerability profiling & threat-intelligence workflow

CVSS | EPSS | CISA KEV | ATTACK | AGE

A capstone project exploring vulnerability profiles beyond severity-only analysis.

THE PROBLEM

Same severity. Very different threat profile.

CVSS is important - but severity alone does not capture exploitation likelihood, known exploitation, attack characteristics or age.

CVE A

CVSS	**9.0**
EPSS	**0.90**
CISA KEV	**YES**
Attack Vector	**NETWORK**

CVE B

CVSS	**9.0**
EPSS	**0.01**
CISA KEV	**NO**
Attack Vector	**LOCAL**

VULNIQ asks: can multiple vulnerability characteristics reveal meaningful natural profiles?

THE ML DECISION

I did not manufacture priority labels.

There was no reliable target such as enterprise remediation priority, so I used unsupervised learning to discover natural vulnerability profiles.

NO	**Artificial Critical / High / Medium / Low labels**	YES	**Unsupervised ML → discover profiles from similarity**

Algorithms evaluated

K-MEANS	Scalable centroid-based clustering
AGGLOMERATIVE	Hierarchical comparison on sample
DBSCAN	Density-based clustering + noise

Selected profiling model: K-Means, K = 4

WHAT WE BUILT

An end-to-end vulnerability profiling pipeline.

Every step below is part of the current VULNIQ capstone.

346,232 CVEs	→ **15** selected features	→ **39** ML-ready features	→ **3** algorithms evaluated	→ **4** vulnerability profiles

FEATURES USED FOR CLUSTERING

Severity & impact • EPSS • CISA KEV • Attack characteristics • Confidentiality / Integrity / Availability impact • Vulnerability age

Preprocessing → feature engineering → encoding → scaling → clustering → evaluation → profiling

Result: four interpretable vulnerability populations - not artificial risk labels.

WHAT THE MODEL FOUND

1.76% → 58.08%

of CVEs → of CISA KEVs

Cluster 0 contained 6,100 CVEs and 960 known exploited vulnerabilities.

CLUSTER 0	**Exploitation Hotspot**	**1.76%**	58.08% KEVs
CLUSTER 1	**High Severity & Impact**	**30.94%**	33.15% KEVs
CLUSTER 2	**Older & Highly Exploitable**	**20.17%**	0% KEVs
CLUSTER 3	**Large Lower-Signal Population**	**47.13%**	8.77% KEVs

KEV was included as a clustering feature; this concentration describes the resulting profile and is not independent predictive validation.

WHAT ENRICHMENT ADDED

From cluster IDs to understandable vulnerability profiles.

After clustering, the CVEs were enriched using a second dataset. CWE, vendor and product did not determine the clusters.

4 PROFILES		CWE		VENDOR		PRODUCT
K-Means output	+	Weakness type	+	Technology maker	+	Affected product

WHAT THIS LET US UNDERSTAND

Weakness patterns	Which CWE categories recur within each profile
Technology concentration	Which vendors and products are common in each profile
Remediation insight	Which vulnerability populations could be analyzed or managed together

Clustering created the profiles. Threat-intelligence enrichment made those profiles easier to interpret.

WHAT VULNIQ ACHIEVED

From 346K vulnerabilities to interpretable profiles.

The project delivers a complete unsupervised ML workflow - from raw vulnerability data to profiled and enriched results.

01	**346,232 CVEs analyzed**	Severity, exploitation, attack and impact characteristics
02	**3 clustering approaches evaluated**	K-Means • Agglomerative • DBSCAN
03	**4 vulnerability profiles identified**	Distinct populations rather than artificial risk labels
04	**Exploitation concentration surfaced**	1.76% of CVEs contained 58.08% of CISA KEVs
05	**Profiles enriched with threat intelligence**	CWE • Vendor • Product

VULNIQ transforms a large vulnerability population into interpretable profiles that can support more focused vulnerability analysis and remediation planning.